

DC MACHINES (15 QUESTIONS)

Conceptual

1. Explain why commutation is necessary in a DC machine and describe what would happen if commutation failed.
2. Discuss the function of the armature winding and explain how its placement affects generated EMF.
3. Why does a DC machine need a field winding, and how does varying field current influence torque production?
4. Describe back EMF in a DC motor and how it stabilizes the motor's operating speed.
5. Explain the purpose of brushes in a DC machine and why brush material selection is important.
6. A DC generator has significant armature reaction. Explain how it distorts the main flux and propose two design methods to reduce this effect.
7. A DC motor is connected to a constant supply but loaded heavily. Discuss the electromagnetic effects that cause the motor to slow down and draw more current.
8. Compare the torque-speed characteristics of a DC shunt motor and a DC series motor and justify their different applications.
9. Explain how interpoles help correct commutation problems and mathematically relate their MMF to armature current.
10. Analyze how iron losses, copper losses, mechanical losses, and stray losses appear differently in motor vs generator operation.
11. Discuss in detail the conditions required for a self-excited DC generator to build up voltage, including the role of residual magnetism.
12. Provide a deep explanation of how load changes affect the voltage regulation of a DC generator and relate this to internal voltage drops.

13. Describe how speed control of a DC motor can be achieved using armature voltage control and field weakening, comparing advantages and disadvantages.
14. Explain the complete torque production mechanism from field excitation to electromagnetic force on the conductors for any DC motor.
15. Assess the operational limitations of DC machines and explain why they are largely replaced by induction and synchronous machines in modern systems.

Workout

1. A DC generator has 4 poles, a no-load flux of 0.021 Wb that drops by 8% at full load, 240 conductors arranged in wave winding, and operates at 980 rpm. Determine the generated EMF at full load. Begin by finding the new flux after the 8% reduction, then convert 980 rpm into revolutions per second. After that, apply the wave-wound EMF equation, showing each substitution and expansion clearly, and leave space to compute the final EMF value.
2. A 220-V DC shunt motor with an armature resistance of $0.28\ \Omega$ and a field resistance of $180\ \Omega$ draws 3.2 A of armature current at no load when running at 1380 rpm. Under load, the armature current increases to 42 A. Determine the back EMF at load and the new speed. First calculate the shunt field current using V/R_f , then compute the back EMF by subtracting $I_a R_a$ from the supply voltage. Next, apply the motor speed relationship that links speed to back EMF for constant flux and use a proportional ratio to determine the new speed, showing all algebraic steps.
3. A 240-V DC shunt motor runs at 1200 rpm while drawing an armature current of 18 A, and its armature resistance is $0.36\ \Omega$. To increase the speed to 1600 rpm, the field must be weakened. Determine the percentage decrease in flux required. Start by calculating the original back EMF, then write the speed–EMF–flux relationship showing that speed is proportional to E/Φ . Use the ratio of initial speed to final speed to establish the flux ratio, rearrange the expression to solve for the percentage flux reduction, and show all symbolic and numerical steps without solving them.
4. A 220-V DC series motor runs at 800 rpm while drawing 12 A and later draws 30 A when the load increases. Find the new speed. Use the relationship that torque is proportional to the square of current and explain how torque, flux and current are related in a series motor. Then discuss the resulting effect on speed, which is inversely related to flux or current depending on the assumed

operating region. Set up the proper ratio and leave space for the numerical substitution and final speed calculation.

5. A self-excited DC shunt generator with a field resistance of $320\ \Omega$ has an open-circuit characteristic showing 1.1 V per ampere of field current, and it also has a residual voltage of 8 V. Determine the final no-load generated voltage. Begin by expressing the field current as $V/320$, then write the open-circuit characteristic equation $E = 8 + 1.1I_f$. At equilibrium, generated voltage equals terminal voltage, so set $E = V$ and write out the algebraic equation. Rearrange it step by step, leaving the space for the calculation of the final value.

6. A 10-kW DC generator delivers its rated load at 230 V. Its losses are: 320 W iron loss, 180 W mechanical loss, 250 W field loss, and armature copper loss given by $I_a^2 R_a$ with an armature resistance of $0.22\ \Omega$. Determine the armature current, armature copper loss, and efficiency. Begin by computing the armature current using P_{out}/V . Then express the copper loss using $I_a^2 R_a$. Afterward sum all losses and add them to the output power to get input power. Finally, write the efficiency formula $P_{out}/P_{in} \times 100\%$, leaving all results unsolved.

7. A 230-V differential compound generator supplies 35 A at full load. The shunt field current is 1.8 A, and the series field reduces the flux by 4% at full load. The no-load EMF is 246 V. Compute the terminal voltage at full load. Start by writing the expression for the reduced flux as 0.96Φ . Then relate EMF to flux and write $E_{g_full} = E_{g_no_load} \times 0.96$. After calculating the generated EMF symbolically, subtract the armature voltage drop $I_a R_a$ (if R_a is not given, note the assumption). Finally, write the expression that yields terminal voltage, leaving blanks for the numerical substitution.

8. A DC machine carrying 52 A in the armature has 360 armature conductors, uses lap winding with 2 poles, and requires the interpole MMF to be 1.15 times the armature MMF per pole. Find the required interpole ampere-turns. Begin by determining the current per path, using I_a divided by the number of parallel paths in lap winding. Compute the number of conductors per pole, multiply by current per path to get armature MMF per pole, and then multiply by 1.15 to find the required interpole MMF. If the same current flows through the interpole coil, express the required turns as MMF/I_a , keeping all values symbolic.

9. A 250-V DC shunt motor with an armature resistance of $0.44\ \Omega$ draws 2.5 A of armature current at no load and runs at 1520 rpm. At rated load, the armature current is 28 A, and the flux decreases by 6% due to armature reaction. Find the speed at full load. First compute the back EMF at no load and at full load using $E = V - I_a R_a$ for each case. Then apply the speed ratio formula $n_2/n_1 = (E_2/E_1)(\Phi_1/\Phi_2)$, substitute all known values step by step, and write the final expression for speed without solving.

10. A 6-pole wave-wound DC generator has 480 conductors, a flux of 0.018 Wb per pole, and rotates at 1000 rpm. It has an armature resistance of $0.32\ \Omega$ and delivers 45 A of load current. Determine the generated EMF, terminal voltage, and voltage regulation. Begin by converting the speed to revolutions per second. Then write the wave-wound armature EMF equation and substitute all known values. After obtaining the symbolic expression for EMF, compute the armature drop $I_a R_a$ and write the terminal voltage expression $V_t = E - I_a R_a$. Finally, write the formula for regulation as $(E - V_t)/V_t \times 100\%$ and leave space for all numerical substitution steps.

SINGLE-PHASE INDUCTION & UNIVERSAL MOTORS

Conceptual

16. Why does a single-phase induction motor produce no starting torque according to the double revolving field theory?
17. Explain how capacitor-start motors create a larger phase shift compared to split-phase motors.
18. Why are shaded-pole motors considered the simplest single-phase motor design?
19. Describe how the centrifugal switch improves performance during motor startup.
20. Explain why PSC motors do not require a centrifugal switch.

21. Compare the torque characteristics of capacitor-start vs capacitor-start–capacitor-run motors and relate this to power factor improvement.
22. Explain how skewed rotor bars reduce magnetic locking and mathematically discuss how this influences torque ripple.
23. Analyze why universal motors can operate on both AC and DC even though armature-reaction effects differ between the two supply types.
24. Discuss the thermal limitations of universal motors during no-load operation and explain why they may overspeed.
25. Examine how the construction differences between single-phase synchronous motors and repulsion motors influence their starting methods.
26. Discuss the complete starting mechanism of a single-phase induction motor, from energizing the windings to reaching 80% synchronous speed.
27. Compare all five types of single-phase induction motors in terms of torque, applications, and internal construction.
28. Describe how laboratory tests (no-load, load, winding resistance, temperature rise) help characterize universal motor performance.
29. Provide a deep explanation of why BLDC motors offer higher efficiency despite requiring electronic commutation.
30. Evaluate the advantages and disadvantages of single-phase induction motors in modern household and commercial applications.

SYNCHRONOUS MACHINES

Conceptual

31. Explain why synchronous generators are considered the primary source of electrical power globally.
32. Compare salient-pole and cylindrical-rotor machines in terms of speed, structure, and application.
33. Why must a DC field current be supplied to the rotor of a synchronous machine?
34. Describe the role of slip rings in rotor excitation and explain the main disadvantage of using them.
35. What is an “infinite bus,” and why is it essential in understanding parallel generator operation?
36. Explain why synchronous motors are not self-starting.
37. Define voltage regulation of a synchronous generator and describe what a negative VR indicates.
38. Discuss armature reaction in a synchronous generator and analyze how it influences terminal voltage under different load power factors.
39. Provide a mathematical explanation of why synchronous generator frequency is locked to rotational speed.
40. Analyze the difference in flux distribution between salient-pole and cylindrical rotors and derive how it affects reactance.
41. Explain how a brushless excitation system eliminates the need for maintenance-heavy slip rings.
42. Examine the condition where a synchronous generator connected to an infinite bus begins to operate as a motor.
43. Derive the relationship between generated EMF, flux, and electrical frequency ($E = K_c \Phi f$) and explain each term's physical meaning.
44. Explain how the short-circuit ratio (SCR) indicates machine stability and relate it to synchronous reactance.

45. Discuss the complete synchronization procedure for connecting a generator to a grid, including voltage, phase sequence, frequency, and phase angle requirements.
46. Evaluate the operational differences between a synchronous generator operating alone versus in parallel with a large power system.
47. Describe how real and reactive power sharing occur between synchronous generators operating in parallel.
48. Provide a detailed explanation of the torque production mechanism in a synchronous motor and the role of the torque angle δ .
49. Discuss the significance of brushless excitation systems and pilot excitors in modern high-power generators.
50. Explain the practical advantages of synchronous motors over induction motors in high-power, low-speed industrial applications.

Workout

1.A 90-kW, 3-phase, Y-connected, 415-V, 50-Hz, cylindrical-rotor synchronous motor operates at rated load with **0.85 power factor lagging**. The synchronous reactance is **3.2 Ω** , and the efficiency excluding field and mechanical losses is **94%**. Calculate:

- (i) **Mechanical power developed,**
- (ii) **Armature current,**
- (iii) **Internal (back) EMF,**
- (iv) **Power angle,**
- (v) **Pull-out torque** if the maximum internal torque occurs at $\delta = 90^\circ$.

Workout format: Begin by computing input power from rated output and efficiency, then calculate line current from three-phase power equation, find the internal EMF using phasor voltage equation ($E = V - jX_s I$), determine δ from the phasor diagram, and finally express pull-out torque using ($P_{\max} = 3EV/X_s$) and ($T = P/\omega_s$), keeping all steps symbolic.

2.

A 110-kW, 3- ϕ , Δ -connected, 480-V, 60-Hz synchronous motor **runs with a 0.9 leading power factor at full load. The synchronous reactance is $2.8\ \Omega$, and the efficiency excluding rotor losses is 96%.**

Calculate:

- (i) Power converted to mechanical form,
- (ii) Line current,
- (iii) Internal generated EMF,
- (iv) Load angle,
- (v) Maximum electromagnetic power **the motor can deliver.**

Workout: Convert Δ voltage to phase voltage, compute input power and current, use per-phase phasor relations to form the EMF expression, solve for the power angle using the real component of power, and calculate peak power from the standard cylindrical-rotor equation.

3.

A 65-kW, 3- ϕ , Y-connected, 400-V, 50-Hz synchronous motor **operates at 0.78 lagging p.f. with an efficiency of 93% (neglecting mechanical losses). The synchronous reactance is $4.1\ \Omega$.**

Determine:

- (i) Developed mechanical power,
- (ii) Stator current,
- (iii) Back EMF,
- (iv) Power angle,
- (v) Pull-out torque at rated speed.

Workout: Start by calculating the three-phase apparent power, then find the magnitude of current. Use the voltage and current phasor relationship across (jX_s) to compute EMF.

With the real power equation ($P = 3VI\cos\phi$), extract the power angle and evaluate torque from ($T = P/\omega$).

4.

A 120-kW, 3- ϕ , 460-V, 60-Hz, Δ -connected synchronous motor **runs at full load with 0.88 leading power factor. The synchronous reactance is $3.6\ \Omega$, and stray losses are neglected so the efficiency is 95%.**

Find:

- (i) Output mechanical power,
- (ii) Line current,
- (iii) Internal EMF,
- (iv) Torque angle,
- (v) Maximum torque capability.

Workout: Derive the phase values for Δ connection, calculate current from input power, express EMF using complex phasor subtraction, compute the angle from the reactive component, and formulate the maximum torque in terms of electrical power converted to torque.

5.

A 55-kW, 3-phase, 415-V, 50-Hz synchronous motor **operates at 0.92 p.f. lagging. The synchronous reactance is $5.0\ \Omega$ and efficiency excluding mechanical losses is 92%.**

Determine:

- (i) Mechanical power developed,
- (ii) Armature current,
- (iii) Internal EMF,
- (iv) Power angle,
- (v) Pull-out power and torque.

Workout: Start with phase voltage conversion, then compute the current from total power, create the phasor equation using Kirchhoff's voltage law, derive δ from the real component of power, and calculate maximum torque from the peak electromagnetic power.

6.

A 150-kW, 3- ϕ , Y-connected, 690-V, 50-Hz synchronous motor **runs at 0.83 p.f. switching from lagging to leading mode depending on field excitation. At the rated load given, assume 0.83 leading, synchronous reactance $4.5\ \Omega$, and efficiency excluding rotor losses 97%.**

Compute:

- (i) Mechanical power developed,
- (ii) Armature current,
- (iii) Internal EMF,
- (iv) Torque angle,

(v) Maximum torque **capability**.

Workout: Compute input power using efficiency, convert to per-phase, determine current and then EMF using complex phasor subtraction, derive the angle via power equation, and determine maximum torque through the standard cylindrical rotor expression.

7.

A 200-kW, 3- ϕ , Δ -connected, 600-V, 60-Hz synchronous motor **operates with** 0.86 lagging power factor at rated load. The synchronous reactance is $3.9\ \Omega$, and efficiency excluding stator copper losses is 96%.

Calculate:

- (i) Mechanical developed power,
- (ii) Line current,
- (iii) Back EMF,
- (iv) Load angle,
- (v) Pull-out power and torque.

Workout: Determine phase current using Δ connection formula, write power equations, express EMF as phasor ($E = V - jX_s I$), find δ using power–angle relationship, and convert peak electromagnetic power to torque.

8.

A 95-kW, 3-phase, Y-connected, 380-V, 50-Hz synchronous motor **delivers rated output at** 0.79 leading. The synchronous reactance is $3.0\ \Omega$, and electrical efficiency (excluding mechanical) is 94%.

Determine:

- (i) Mechanical power output,
- (ii) Armature current,
- (iii) Internal EMF,
- (iv) Power (torque) angle,
- (v) Maximum torque at rated frequency.

Workout: Use power triangle and input power to calculate current, express EMF from phasor difference, derive δ using real power expression, and calculate peak torque using the synchronous torque–power relation.

9.

A 140-kW, 3- ϕ , 500-V, 60-Hz synchronous motor **operates fully loaded at 0.82 lagging with an efficiency (excluding field losses) of 95.5%. The synchronous reactance is 4.2 Ω .**

Find:

- (i) Power converted to mechanical output,
- (ii) Line current,
- (iii) Back EMF,
- (iv) Power angle,
- (v) Pull-out torque.

Workout: Compute input using efficiency, obtain current from three-phase power equation, derive EMF using per-phase phasor subtraction, calculate δ from the active power equation, and use the maximum power formula ($P_{\max} = 3EV/X_s$) to determine torque.
10.

A 180-kW, 3-phase, Y-connected, 660-V, 50-Hz synchronous motor **achieves rated output with 0.87 leading p.f. The synchronous reactance is 5.2 Ω , and the efficiency excluding stator iron losses is 96%.**

Compute:

- (i) Mechanical developed power,
- (ii) Armature current,
- (iii) Internal generated EMF,
- (iv) Torque (load) angle,
- (v) Maximum electromagnetic torque.

Workout: Begin by evaluating the total input power, compute current from three-phase equations, write the complex phasor difference between terminal voltage and reactance drop to obtain EMF, derive δ by comparing active power components, and determine peak torque using the synchronous motor maximum power formula.